Urban Livability and Pet Infrastructure: A Global Analysis of Dog Park Accessibility, Population Aging, and Environmental Indicators

1. Introduction

The current global urbanization and demographic shift to an aging society are changing the expectations of urban recreation infrastructure. Specialized amenities have become an important component in determining urban livability, with emotional benefits that go beyond parks, such as accommodating the diverse and changing needs of city dwellers (Wolch et al., 2014). A case in point is dog parks. Typically, these are perceived to be an incidental form of recreation, but they are also places where people can socialize, stay active, and create community (Urbanik & Morgan, 2013), which is particularly relevant in the context of a growing elderly population. Many seniors depend on their pets to help maintain their physical activity and combat social isolation (Buffel et al., 2012).

While there is has been a growing interest among academics in understanding the connection between urban green spaces and public health (Kondo et al., 2018), there have not been many empirical studies examining the global availability of pet-specific infrastructure across countries. This study contributes to filling this gap by examining the distribution of dog parks by country within a broader context of demographic, economic, and environmental factors. The main hypothesis is that the availability of dog parks is affected, not only by the wealth of a country but by multiple demographic, cultural, and urban planning factors, especially the age of the population and the cultural attitudes towards pet ownership.

2. Research Objectives

- To measure the place of dogs parks in each country by looking at how many there are compared to the population of that country (dogs parks per 100, 000 people).
- To see if there are differences in how many dog parks there are per person based on three other types of data (economic development; public health) and one other type of difference (environmental quality).
- To identify countries that are similar in terms of having the same dog park availability and to then create a cluster analysis of countries that have the same aging / elderly population and dog park availability using geographically weighted regression (GWR) for an individual and the country as a whole.

3. Data and Methodology

3.1 Data Sources

To build a comparative dataset that compares multiple dimensions through integrating geospatial data and socioeconomic indicators. Dog parks were sourced from OpenStreetMap (OSM) data through the Overpass API which provides dog park features labelled with “leisure=dog_park,” from all over the world. OSM data is widely regarded as one of the largest sources of volunteered geographic information (VGI) available globally, yet there are many differences across the globe (for example= how much of the globe has OSM data, what geographies have greater or lesser OSM data) and data quality is directly correlated to the density of local contributors (Haklay, 2010). The limitations of using OSM are included below in the constraints section to outline issues related to OSM and data limitations.

To provide socioeconomic context, 18 indicators were retrieved from the World Bank API, organized into five thematic domains:

- **Demographic Dynamics:** Population size, growth rate, density, and urbanization levels.

- **Age Structure:** Elderly ratio (65+), youth ratio (0–14), life expectancy, death rate, and the aging index.
- **Health and Economy:** GDP per capita, GINI index, health expenditure per capita, and hospital bed availability.
- **Environmental Context:** Forest cover, carbon emissions per capita, and agricultural land use.

3.2 Spatial Data Management

The processing and analyzing of all spatial data are done in R. Country-level boundary data are imported via the sf package, which provides a standardized way to handle vector geometries and coordinate reference systems. The tmap and ggplot2 packages are used to create choropleth maps displaying the global distribution of dog park density. The maps are designed in accordance with established principles of cartographic communication, with an emphasis on classification schemes, color palettes, and legend construction to reduce the visual bias in the representation of spatial inequality.

3.3 Analytical Pipeline

There are four sequential phases in the analytical framework. The first involves performing exploratory spatial analysis to identify the distribution of population within the region and quantify the intra-regional inequality with Gini coefficients. In the second phase, spatial autocorrelation of the spatial distributions is conducted using Global Moran's I and Local Indicators of Spatial Autocorrelation (LISA) with the spdep package. Additionally, the Getis-Ord Gi* statistic is used to identify infrastructure concentrations. Thirdly, k-means clustering (k=4) will be completed for each country utilizing demographic, economic and environmental variables to create country profiles. Finally, a Geographically Weighted Regression model created using the spgwr package will be used to determine how demographic aging and park density are related in a spatial context.

3.4 Interactive Visualization

An interactive website dashboard that is created using HTML, CSS and JavaScript (with the use of Leaflet.js for map rendering and Chart.js to create statistical visualizations) was developed so users can access the data they seek through a web-based interface. This allows users to search for the number of dog parks (the density), examine different demographic indicators and analyse cluster types based on country.

References

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